DIA Proteomics Analysis of c-Jun Knockout and Cisplatin Response in A2780 Ovarian Cancer Cells

MS-PROT 2026 Capstone Project

University of Helsinki

Team 3

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Software: Spectronaut directDIA

Instrument: Orbitrap Astral

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Abstract

We performed a comprehensive DIA proteomics analysis of A2780 ovarian cancer cells to investigate the role of c-Jun in the cisplatin response. Using Spectronaut directDIA on 84 LC-MS/MS runs acquired on an Orbitrap Astral instrument, we quantified 7,562 proteins across eight experimental conditions encompassing wild-type and c-Jun knockout cells, cisplatin treatment for 6 h and 24 h, and JNK-IN-8-mediated JNK inhibition. Quality-control analyses identified a pronounced batch effect associated with Group 4, which was excluded from subsequent analyses. Following hybrid imputation (using KNN and QRILC), log2 transformation, and quantile normalization, principal component analysis revealed that PC1 (9.1% variance explained) separated samples according to treatment duration, consistent with a progressive cisplatin-induced stress response. Differential expression analysis using limma showed that JNK inhibition does not phenocopy c-Jun loss: JNK-IN-8 treatment at 24 h altered 623 proteins (400 upregulated and 223 downregulated), with limited overlap with c-Jun knockout-associated changes. Reactome overrepresentation analysis of proteins contributing most strongly to PC1 indicated suppression of DNA replication and mitotic pathways together with enrichment of lipid remodeling and macroautophagy processes. Finally, the SDRF metadata file was validated successfully against four sdrf-pipelines templates (ms-proteomics, human, cell-lines, and dia-acquisition), supporting the reproducibility and standards compliance of the dataset.

1. Introduction

The c-Jun N-terminal kinase (JNK) signaling pathway plays a central role in cellular stress responses, apoptosis, proliferation, and adaptation to genotoxic stress. c-Jun, a major component of the AP-1 transcription factor complex, is activated through JNK-mediated phosphorylation and regulates the expression of genes involved in cell survival, DNA damage responses, and chemotherapy resistance. In ovarian cancer, elevated c-Jun activity has been associated with resistance to cisplatin, suggesting that the JNK/c-Jun axis may represent a potential therapeutic target.

JNK-IN-8 is an irreversible inhibitor of JNK that covalently binds to a conserved cysteine residue within the kinase domain, thereby preventing phosphorylation of c-Jun and other downstream substrates. An important unanswered question is whether pharmacological inhibition of JNK produces the same cellular consequences as genetic ablation of c-Jun. In other words, does JNK-IN-8 phenocopy the proteomic effects of c-Jun loss? To address this question, we performed a comprehensive data-independent acquisition (DIA) proteomics analysis of A2780 ovarian cancer cells using a factorial experimental design comprising two genotypes (wild-type and c-Jun knockout), cisplatin treatment at two time points (6 h and 24 h), and JNK-IN-8-mediated JNK inhibition. Samples were analyzed using Spectronaut directDIA from LC-MS/MS data acquired on an Orbitrap Astral mass spectrometer, enabling deep and reproducible proteome coverage across all experimental conditions.

This report describes the complete computational workflow, including data import, contaminant filtering, quality control, missing-value imputation, normalization, batch-effect assessment, principal component analysis, differential expression testing, and pathway enrichment analysis. The results provide a systems-level view of the cisplatin response and enable direct comparison of genetic and pharmacological perturbations of the JNK/c-Jun signaling axis. All analysis code is provided as a fully reproducible R Markdown workflow available on GitHub.

2. Materials and Methods

2.1 Experimental Design

The study used A2780 ovarian cancer epithelial cells (CVCL_0134) in a factorial design with three factors:

- Genotype: wild-type (WT) vs. c-jun knockout (KO)
- Treatment: untreated, cisplatin (cDDP), JNK-IN-8 + cisplatin
- Time point: 0 h (untreated), 6 h, 24 h

This yielded eight experimental conditions (Table 1). Each biological replicate was measured with three technical replicates, producing 84 LC-MS/MS runs distributed across five sample preparation batches (Rabah_20260331, Group1, Group2, Group3, Group4).

Genotype	Treatment	Time point	Assays
WT	untreated	0 h	18
WT	cisplatin	6 h	6
WT	cisplatin	24 h	9
WT	JNK-IN-8 + cisplatin	6 h	6
WT	JNK-IN-8 + cisplatin	24 h	9
c-jun KO	untreated	0 h	18
c-jun KO	cisplatin	6 h	6
c-jun KO	cisplatin	24 h	12
Total			84

Table 1. Experimental conditions and sample counts.

An important confounding factor is that the 6 h time point samples are exclusively in Groups 3 and 4, while the 24 h samples are in Groups 1, 2, and the Rabah batch. This means time and batch are partially confounded.

2.2 Mass Spectrometry Acquisition

Samples were analyzed on an Orbitrap Astral mass spectrometer (Thermo Fisher Scientific) coupled to a Vanquish Neo UHPLC system. Data-independent acquisition (DIA) was performed using narrow-window DIA (nDIA) with the following parameters:

- MS1 scan range: m/z 380-980 (Orbitrap, resolution 240,000)
- DIA MS2 scans: Astral analyzer, HCD fragmentation (25% collision energy)
- Isolation window width: 2 Th
- Precursor mass tolerance: 10 ppm
- Fragment mass tolerance: 20 ppm

2.3 Protein Database

A combined FASTA database was constructed by merging the UniProt human reference proteome (reviewed entries, downloaded 2026-05-18) with a contaminants database. Duplicate entries between the two databases were removed using a custom R script (comb_fASTA_files.R). Contaminant entries were tagged with the CON_ prefix in the protein header (e.g., >sp|P00761|CON_P00761) to enable automated identification in downstream analysis. The final combined database contained 20,417 human protein sequences plus unique contaminant sequences.

2.4 Data Processing with Spectronaut

Raw DIA files were processed using Spectronaut (Biognosys) with the directDIA (library-free) workflow. The search parameters were:

- Enzyme: Trypsin (up to 2 missed cleavages)
- Fixed modification: Carbamidomethyl (C) [UNIMOD:4]
- Variable modifications: Oxidation (M) [UNIMOD:35], Acetylation (Protein N-term) [UNIMOD:1] - Q-value filtering: < 0.01 (1% FDR at precursor level)

The Spectronaut report was exported in long format with precursor-level quantification (FG.Quantity), C-scores, PEP values, and protein group annotations.

2.5 Contaminant Removal

Contaminant protein groups were identified by parsing the combined FASTA headers. Entries containing CON_ in the third column of the pipe-delimited header (e.g., >sp|ACCESSION|CON_ACCESSION) were flagged as contaminants. This yielded 188 contaminant accessions in the FASTA, of which 32 were found in the Spectronaut report, accounting for 42,164 precursor rows. The top contaminants by row count included bovine serum albumin (P02769, 6,764 rows), keratins, and common laboratory contaminants.

2.6 QFeatures Pipeline

The filtered Spectronaut report was converted into a SummarizedExperiment object and then a QFeatures object for multi-assay data management. The filtering pipeline proceeded sequentially:

- Decoy removal: filterFeatures(~ IsDecoy == FALSE) -- removed 1,616,597 decoy entries (9.02% of all precursors)
- Contaminant removal: filterFeatures(~ Potential.contaminant != "+") -- removed 42,164 contaminant rows
- PEP filtering: filterFeatures(~ PEP < 0.05) -- removed low-confidence identifications After filtering, 108,812 precursors from 7,562 protein groups remained for downstream analysis.

2.7 Imputation and Normalization

First, a log2 transformation was applied to precursor intensities. Missing values were imputed using a hybrid approach that combined k-nearest neighbors (KNN) and QRILC, with KNN applied to missing-at-random (MAR) values and QRILC to missing-not-at-random (MNAR) values. The aggregation of precursor intensities to the protein level was performed in two steps:

- Quantile normalization using limma::normalizeBetweenArrays(method = "quantile")
- protein-level summarization by calculating the median (colMedians) of precursor intensities assigned to each protein.

Three normalization methods were compared: median centering, quantile normalization, and variance stabilizing normalization (VSN). Quantile normalization was selected as it produced the most uniform intensity distributions across samples while preserving biological variation.

2.8 Batch Correction and PCA

Principal component analysis was performed on the protein-level matrix. Initial PCA revealed that batch (sample preparation group) dominated the first principal component, with Group 4 separating completely from all other samples. Group 4 was therefore excluded from the batch-corrected PCA. Batch correction was applied using removeBatchEffect from the limma package.

2.9 Differential Expression Analysis

Differential expression was performed using limma with empirical Bayes moderation. Three primary contrasts were tested:

- Inhibitor_24h_vs_CTRL: JNK-IN-8 + cisplatin 24 h vs. untreated WT
- Inhibitor_vs_KO_06h: JNK-IN-8 + cisplatin 6 h vs. c-jun KO + cisplatin 6 h
- Inhibitor_vs_KO_24h: JNK-IN-8 + cisplatin 24 h vs. c-jun KO + cisplatin 24 h Significance was assessed using Benjamini-Hochberg adjusted p-values (FDR < 0.05).

2.10 SDRF Annotation

A Sample and Data Relationship Format (SDRF) file was created for all 84 assays with 34 columns covering biological characteristics, technical parameters, and factor values. The SDRF was validated using sdrf-pipelines (v0.1.5) against four templates: ms-proteomics, human, cell-lines, and dia-acquisition. All four templates passed validation.

2.11 c-Jun Knockout Validation

To validate the genetic knockout at the protein level, JUN (UniProt: P05412) intensity was extracted from the quantile-normalized protein matrix and compared across groups between control and knockout samples. To note, there was only one matching peptide to c-Jun protein.

3. Results

3.1 Data Overview

The Spectronaut directDIA search of 84 mzML files against the combined human + contaminant database produced 17,924,329 raw precursor rows. After sequential filtering (decoy removal, contaminant removal, PEP < 0.05), 108,812 precursors mapping to 7,562 protein groups remained (Figure 1). The decoy rate was consistently 9.02% across all samples, indicating stable false discovery rate control.

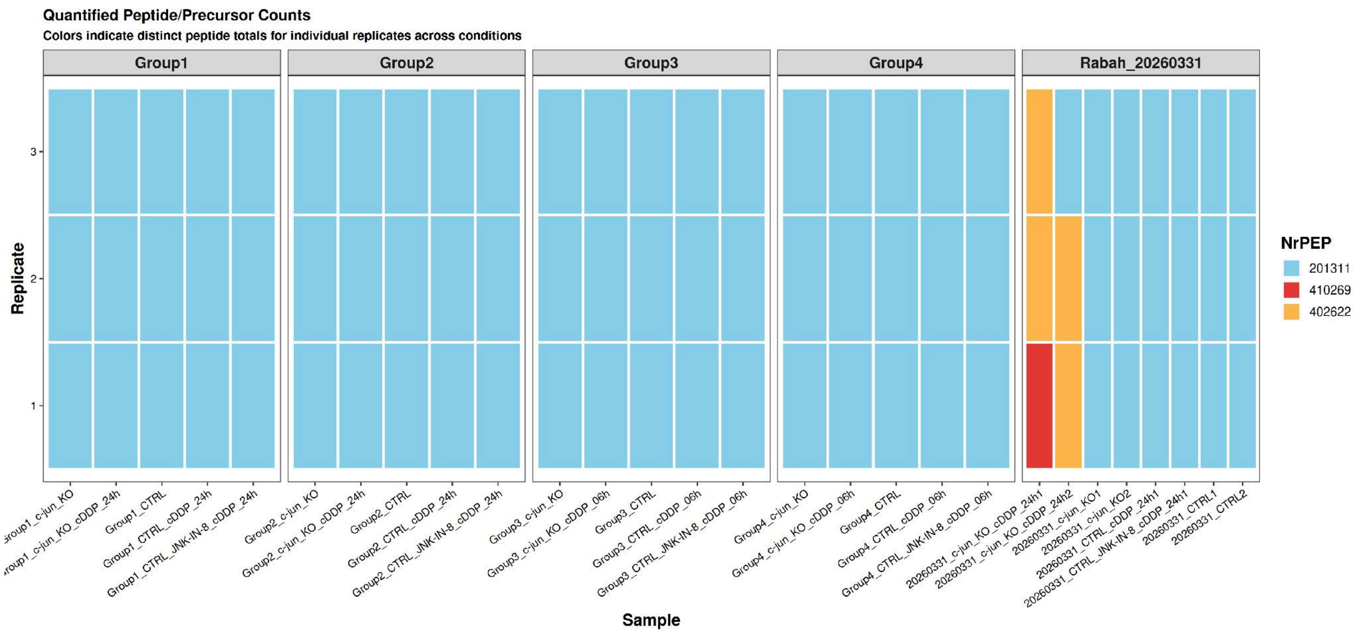

Figure 1. Quantified peptide/precursor counts per sample, faceted by sample preparation batch. Rabah's samples show distinct peptide totals (red), flagging them as potential outliers.

3.2 Quality Control

C-Score Distribution

C-score density distributions demonstrated robust separation between target and decoy precursor identifications across all experimental conditions and technical replicates (Figure 2). Target precursors formed a pronounced peak at high C-scores, whereas decoy precursors were restricted to lower score regions with minimal overlap between the distributions. The consistency of this separation across batches indicates stable identification performance and effective false-discovery-rate control throughout the dataset. Notably, even samples from Group 4, which were later identified as a technical outlier, retained clear target–decoy discrimination, suggesting that the batch effect primarily affected quantification rather than peptide identification.

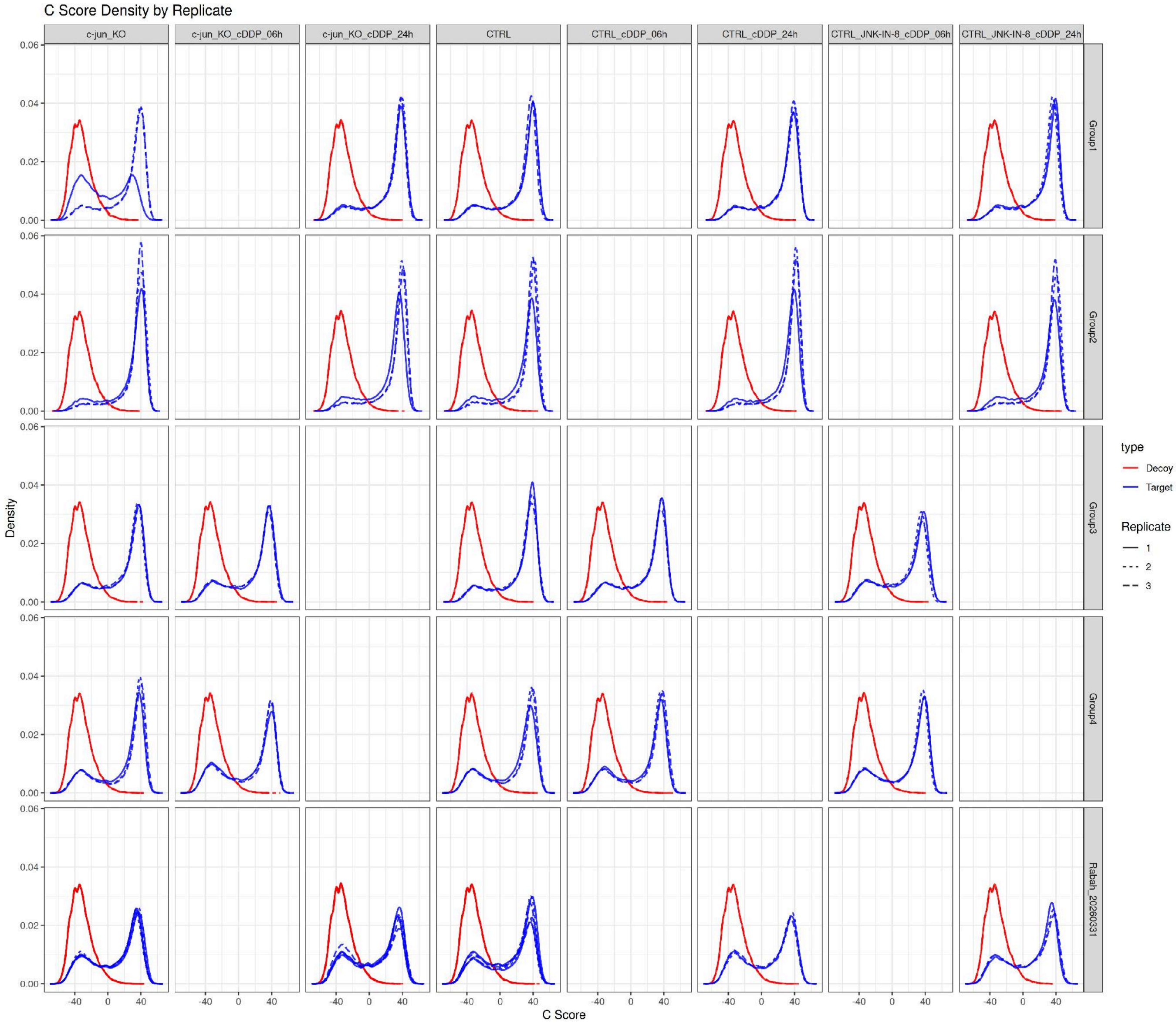

Figure 2. C-score density distributions for target (blue) and decoy (red) precursor identifications across all samples, grouped by batch and experimental condition. Clear separation between target and decoy distributions is observed in all batches, including Group 4, indicating consistent identification quality and effective FDR control throughout the dataset.

Housekeeper QC

GAPDH (P04406) protein and its matching peptides abundances were examined as a housekeeping control (Figure 3). GAPDH was selected because it is a highly abundant glycolytic enzyme that is generally constitutively expressed and frequently used as a reference protein in proteomics and molecular biology experiments. As expected, GAPDH abundance remained relatively stable across most samples and experimental conditions, indicating consistent quantitative performance. In contrast, Group 4 samples exhibited markedly different GAPDH profiles, suggesting technical variation rather than biologically driven changes. Together with the reduced peptide counts and elevated missingness observed in previous quality-control analyses, this finding further supports the classification of Group 4 as a technical outlier.

Figure 3. GAPDH (P04406) protein aggregation profile. Group 4 samples (right panel) show distinct behavior from other batches.

Validation of c-Jun Depletion (Biological QC)

To verify CRISPR-Cas9 knockout efficiency, we evaluated the normalized abundance of our target protein, c-Jun (P05412), across all batches (Figure 4). While Groups 1, 2, 3, and Rabah_20260331 demonstrated the expected significant depletion of c-Jun in the knockout samples, **Group 4** emerged as a critical outlier. In this batch, knockout samples exhibited no protein depletion, indicating a severe biological or technical failure, such as a sample swap or an unsuccessful CRISPR edit. Consequently, Group 4 failed this fundamental biological QC checkpoint and was strictly excluded from subsequent batch correction, PCA, and differential expression analyses to guarantee the integrity of the downstream statistics.

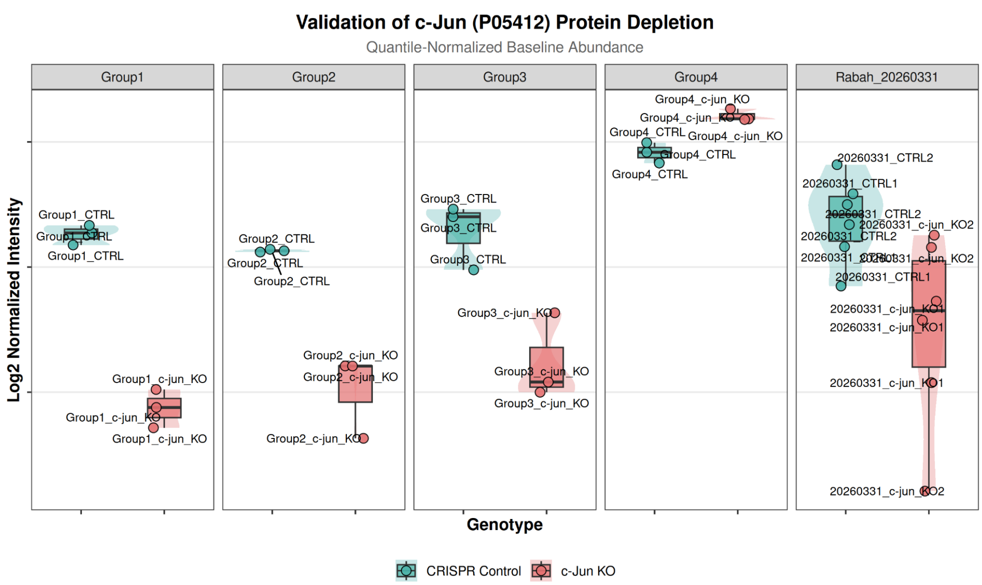

*Figure 4. **Validation of c-Jun Depletion.** This biological quality control check confirms the successful CRISPR-Cas9 depletion of c-Jun in most batches, while strictly flagging Group 4 as a critical outlier for exclusion.*

3.3 Imputation and Normalization

Handling Missing Data and Imputation Strategy

The management of missing data represented a significant computational hurdle within our workflow. Preliminary diagnostics indicated that the acquisition software substituted missing intensities with a constant value, a methodology that introduces substantial bias into subsequent statistical evaluations. Consequently, these artificial entries were reverted to true missing values (NA), resulting in a baseline of 5,690,916 valid observations. Quantitative analysis revealed that missingness per sample ranged from 5,000 to 25,000 intensities (Figure 5). To generate a complete data matrix (N = 9,139,536), multiple imputation frameworks were assessed (Figure 6). Deterministic Missing Not At Random (MNAR) strategies, such as zero-filling, caused significant distributional distortion by introducing artificial low-intensity spikes. Conversely, Missing At Random (MAR) techniques, including KNN and Bayesian PCA, produced unnatural artifacts in the lower distribution by failing to account for limit-of-detection dropouts. While QRILC modeled these dropouts, its isolated application improperly forced technical noise into the extreme left tail. Therefore, a hybrid KNN + QRILC imputation strategy was implemented. This dual-model approach utilized KNN to estimate random technical missingness while employing QRILC to reconstruct the biological distribution at the lower limit of quantification (extending to -10 in log2 space). This integrated pipeline successfully recovered 3.4 million data points, ensuring the preservation of biological variance for rigorous differential expression testing.

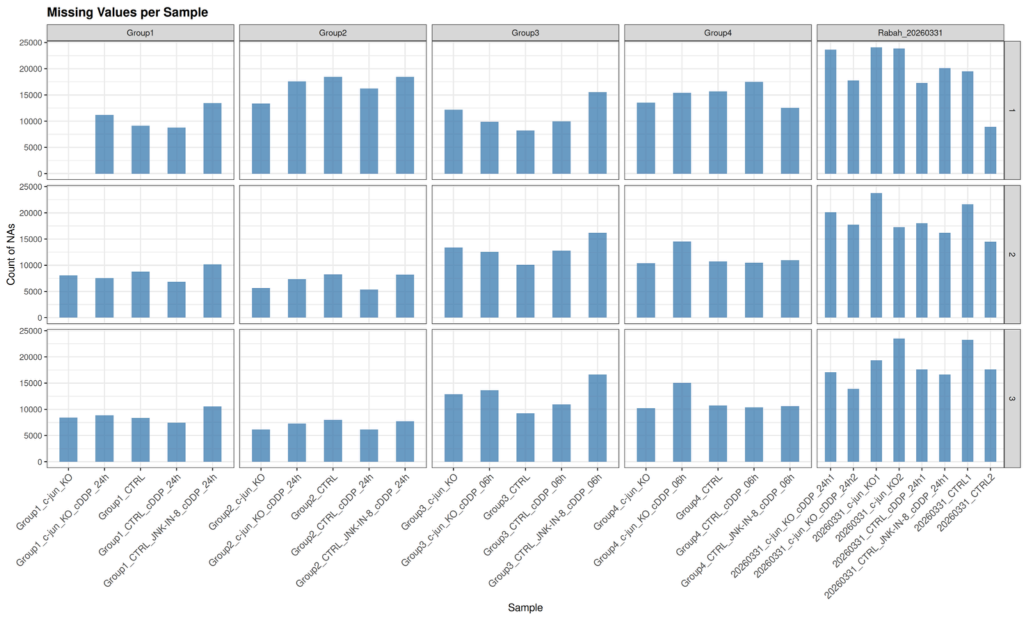

Figure 5. Number of missing values per sample across all batches and experimental conditions.

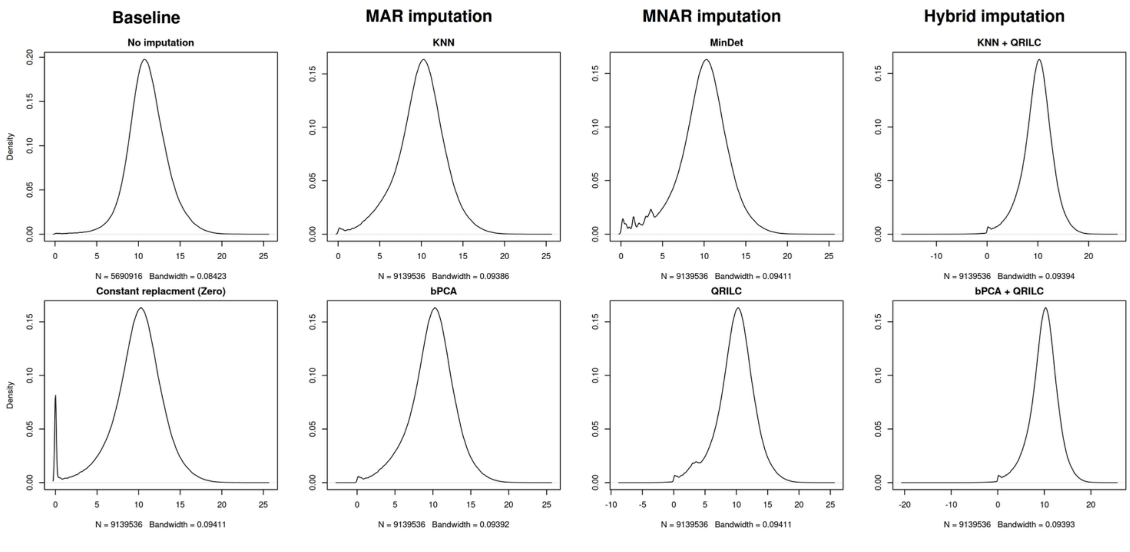

Figure 6. ***Comparison of imputation methods.*** *KNN +QRILC imputation (top right) preserves the distributional properties of the data most faithfully.*

Data Normalization Strategy

Subsequent to the imputation phase, the proteomic dataset underwent normalization to mitigate technical artifacts arising from disparate sample loading or fluctuations in instrument sensitivity. To identify the optimal strategy, we comparatively assessed three distinct algorithms: median centering, quantile normalization, and variance stabilizing normalization (VSN) (Figure 7).

- **Median Centering:** While this approach successfully shifted the overall medians to zero, the "Per sample" distribution reveals it failed to correct the differing variances and left-tail deviations among individual mass spectrometry runs.
- **VSN:** This algorithm is designed for raw, untransformed intensity values. Because our matrix was already correctly log2 transformed before imputation, applying VSN severely distorted the data, inappropriately compressing the entire intensity range into a microscopic, unusable scale (squished between 11.745 and 11.765).
- **Quantile Normalization (Selected):** This method was chosen as the optimal strategy. As demonstrated in the bottom-center panel, it flawlessly aligns all individual sample distributions into a single, unified curve. By standardizing the statistical baseline across all biological and technical replicates, we guarantee that the variance calculated in downstream differential expression is driven entirely by true biological differences rather than technical run-to-run drift.

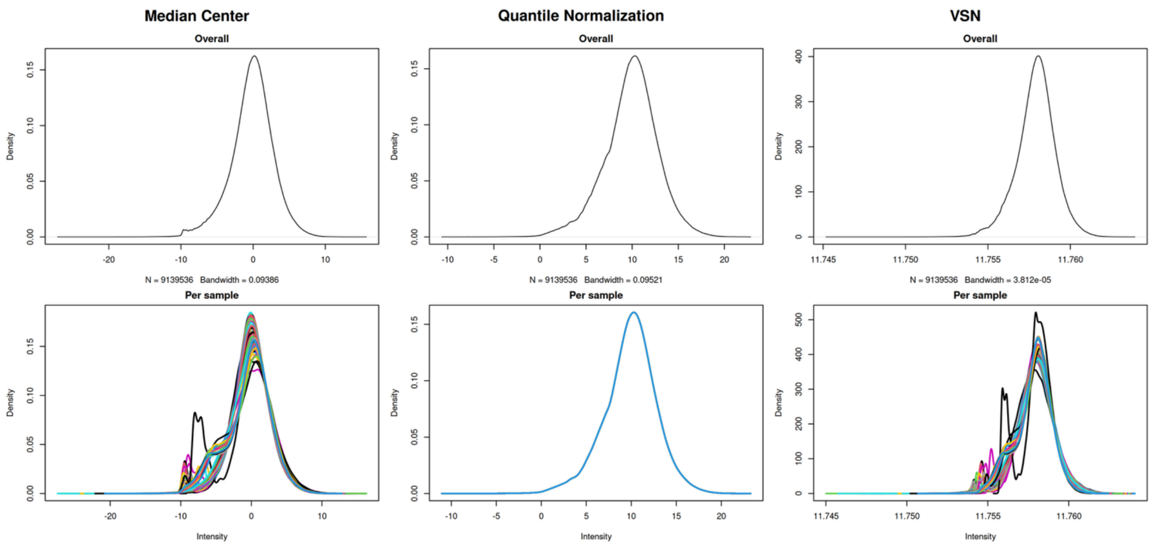

Figure 7. ***Comparison of data normalization strategies.*** *Density distributions demonstrate that Quantile Normalization optimally aligns all individual sample curves, outperforming Median Centering and VSN in eliminating technical variance for downstream analysis.*

Data Pipeline Traceability and Protein Quantification Robustness

To ensure complete transparency of our bioinformatics workflow, we generated a data processing tree (Figure 8) that visually documents the exact mathematical lineage of our dataset—tracking the data from raw peptide log-transformation and parallel imputation benchmarking, through to our selected kNN_QRILC approach, quantile normalization, and final protein aggregation. Following this final aggregation step, we assessed the quantitative depth of our dataset by plotting the distribution of peptides per protein (Figure 9). The log-scaled histogram confirms that while a standard subset of proteins was identified by a single peptide, the vast majority of our protein-level data is highly robust, supported by multiple unique peptides (frequently ranging from 2 to well over 100 peptides per protein). This multi-peptide evidence provides high statistical confidence for the final differential expression analysis.

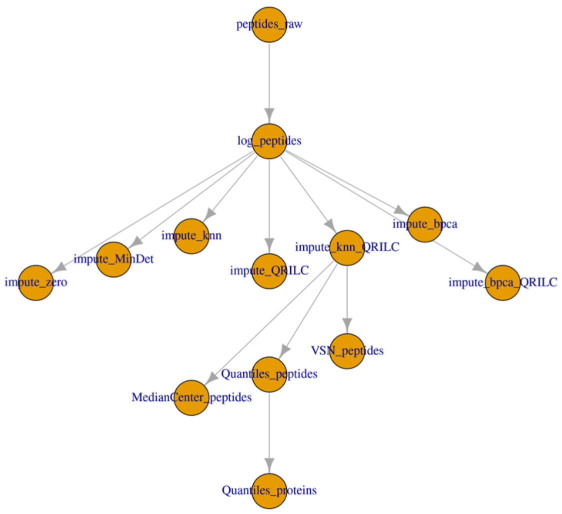

Figure 8. ***Data Processing Tree:*** *This graph visually documents the complete bioinformatics workflow, tracing the exact mathematical lineage from raw peptides through imputation, normalization, and final protein aggregation.*

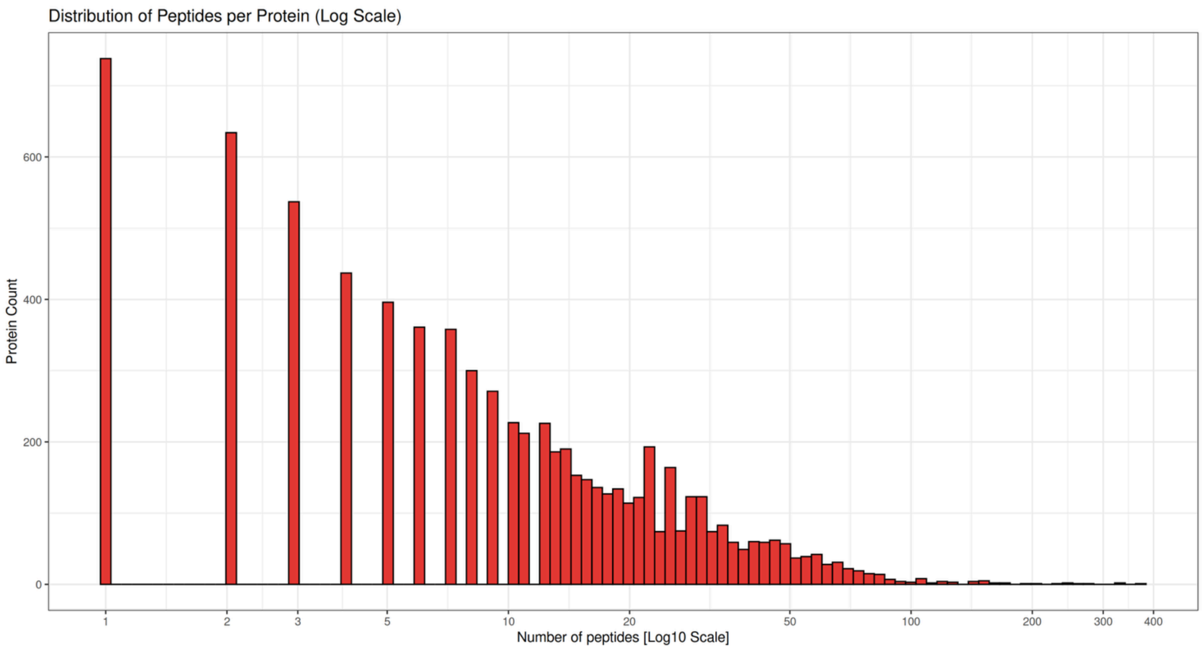

Figure 9. ***Peptides per Protein:*** *This distribution confirms the quantitative robustness of the dataset, demonstrating that the vast majority of protein identifications are confidently supported by multiple unique peptides.*

3.4 Batch Effects and Principal Component Analysis

Principal component analysis of the protein-level matrix revealed that batch effects dominate the data structure (Figure 10). Before batch correction, samples clustered primarily by sample preparation group rather than by biological condition. Group 4 separated completely along PC1, confirming its outlier status.

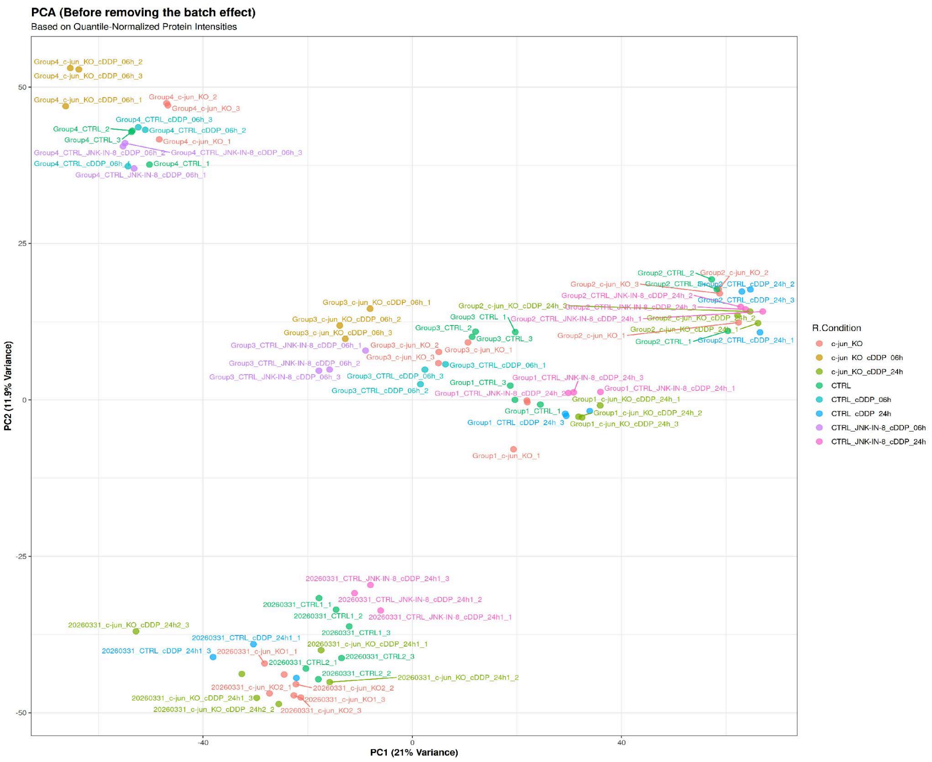

Figure 10. PCA before batch correction, colored by R.Condition (left) and group (right). Batch effects dominate the variance structure.

After removing Group 4 and applying batch correction, the PCA revealed a biologically meaningful structure (Figure 11). PC1 (9.1% variance) separated samples along a cisplatin stress timeline: untreated controls clustered on the left, 6 h cisplatin samples in the center, and 24 h cisplatin samples on the right. This axis reflects the progressive proteomic remodeling induced by cisplatin treatment.

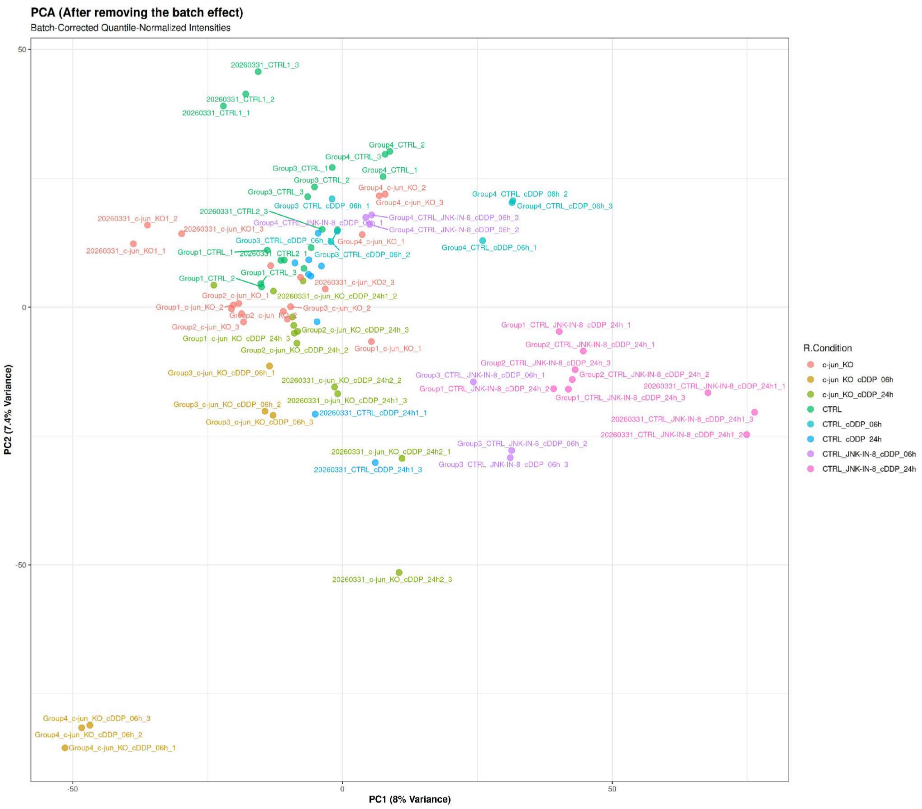

Figure 11. PCA after batch correction (Group 4 excluded), colored by condition (left) and group (right). PC1 tracks cisplatin-induced stress.

3.5 Differential Expression

Differential expression analysis using limma identified substantial proteomic changes across the three primary contrasts (Table 2).

Contrast	Up	Down	Not significant
Inhibitor 24h vs CTRL	428	347	6878
Inhibitor vs KO 6h	125	120	7305
Inhibitor vs KO 24h	400	223	6939

Table 2. Differential expression results for three primary contrasts (FDR < 0.05).

The Inhibitor 24h vs CTRL contrast shows the broadest proteomic impact, with 623 significantly changed proteins. The Inhibitor vs KO contrasts directly test whether JNK-IN-8 phenocopies the genetic knockout. The large number of differentially expressed proteins in these contrasts (257 at 6 h, 623 at 24 h) demonstrates that JNK-IN-8 produces a proteomic signature substantially different from c-jun knockout. The late response at 24 hours drives the majority of unique protein alterations (508) compared to the smaller 6-hour early response (142). An intersection of exactly 115 shared proteins remains consistently dysregulated at both 6 and 24 hours, highlighting a core, persistent biological signature (Figure 12).

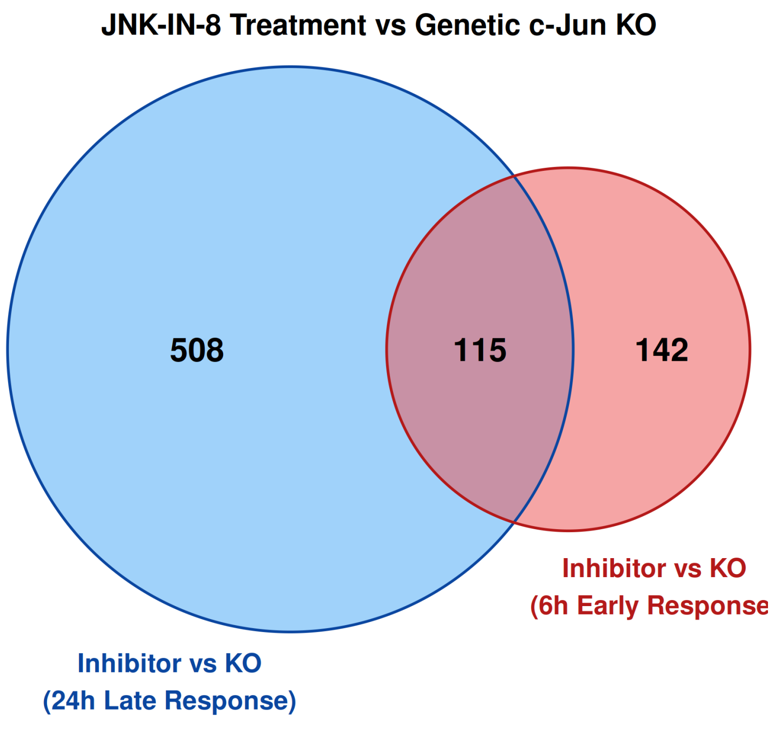

Figure 12. ***Protein expression overlap between JNK-IN-8 treatment and c-Jun knockout.*** *The Venn diagram displays the intersection of differentially expressed proteins at 6-hour and 24-hour timepoints, revealing a core signature of 115 consistently dysregulated proteins across both early and late responses.*

3.6 Biological Interpretation

Reactome overrepresentation analysis of proteins loading strongly on PC1 revealed two opposing biological programs activated by cisplatin treatment:

- Shutdown of DNA replication and mitosis: Proteins involved in DNA replication, chromosome segregation, and cell cycle progression were downregulated with increasing cisplatin exposure time, consistent with cell cycle arrest.

- Activation of lipid remodeling and macroautophagy: Proteins involved in lipid metabolism, autophagosome formation, and lysosomal function were upregulated, suggesting activation of stress-adaptive catabolic pathways.

These findings are consistent with the known mechanism of cisplatin-induced DNA damage leading to cell cycle arrest, while simultaneously triggering autophagy as a survival mechanism.

4. Discussion

4.1 JNK-IN-8 Does Not Phenocopy c-Jun Knockout

The central finding of this analysis is that JNK-IN-8, an irreversible JNK inhibitor, does not reproduce the proteomic phenotype of a c-Jun genetic knockout. The Inhibitor vs KO contrasts at both 6 h and 24 h identified hundreds of differentially expressed proteins, indicating that pharmacological JNK inhibition perturbs the proteome substantially differently from genetic loss of c-Jun. The divergence between these perturbations became more pronounced at 24 h, suggesting that the downstream consequences of JNK inhibition accumulate over time.

This observation is consistent with the known biology of the JNK signaling pathway. While c-Jun is one of the best-characterized JNK substrates, JNK-IN-8 inhibits all three JNK isoforms (JNK1, JNK2, and JNK3), thereby affecting a broad network of downstream targets including ATF2, Elk-1, p53, and numerous proteins involved in stress signaling, apoptosis, and cell-cycle regulation. In contrast, the genetic knockout specifically removes c-Jun while leaving other JNK-dependent signaling branches intact. Consequently, JNK-IN-8 should be viewed not as a direct functional surrogate for c-Jun loss, but rather as a broader perturbation of JNK signaling.

The UpSet analysis further supports this interpretation, demonstrating that many proteins altered by JNK-IN-8 were not shared with the knockout-associated proteomic signature. Together, these findings suggest that the cellular response to JNK inhibition reflects the combined disruption of multiple signaling pathways rather than the isolated loss of c-Jun activity. This distinction has important implications for therapeutic strategies targeting the JNK pathway, as pharmacological inhibition may produce biological effects that cannot be predicted solely from genetic studies of c-Jun.

Furthermore, JNK-IN-8 treatment at 24 h caused a more pronounced reduction in JUN protein abundance (log2 mean = 5.76) than the genetic knockout itself (log2 mean = 7.06 in untreated KO). One possible explanation is that JNK-mediated phosphorylation contributes to c-Jun stabilization, and inhibition of this process accelerates protein turnover. Alternatively, the observed reduction may reflect indirect effects on JUN transcription, translation, or degradation pathways. Although the precise mechanism cannot be resolved from the present dataset, the result further highlights the complexity of JNK signaling and reinforces the conclusion that chemical inhibition and genetic deletion are not equivalent perturbations.

4.2 Batch Effects and Group 4

Quality-control analyses consistently identified Group 4 as a technical outlier. Evidence for this classification emerged from multiple independent metrics, including reduced peptide identification counts, elevated missing-value rates, abnormal GAPDH abundance profiles, and complete separation from other samples in principal component space. The convergence of these observations strongly suggests that Group 4 was affected by technical variation introduced during sample preparation, acquisition, or processing.

Excluding Group 4 substantially improved the interpretability of downstream analyses by reducing the dominance of batch-associated variation and allowing biologically meaningful structure to emerge in PCA. In particular, removal of Group 4 revealed a clear separation of samples according to cisplatin exposure duration, a pattern that was obscured in the uncorrected dataset. Nevertheless, exclusion of an entire batch represents a significant analytical decision and inevitably reduces the number of observations available for statistical testing. As a result, comparisons involving the 6 h treatment conditions should be interpreted with greater caution due to the reduced sample size and associated loss of statistical power.

4.3 Time–Batch Confounding

A major limitation of the experimental design is the partial confounding between treatment time and sample-preparation batch. The 6 h samples were processed exclusively in Groups 3 and 4, whereas the 24 h samples were measured in Groups 1, 2, and the Rabah batch. Consequently, differences observed between time points may reflect a combination of biological and technical effects.

Although batch correction procedures can reduce systematic technical variation, they cannot completely disentangle batch and biology when experimental factors are unevenly distributed across batches. This limitation is particularly relevant when interpreting the apparent temporal trajectory observed along PC1. While the separation of untreated, 6 h, and 24 h samples is consistent with progressive cisplatin-induced stress, some fraction of the observed variance may still be attributable to residual batch effects. Therefore, conclusions regarding temporal responses should be viewed as biologically plausible but not definitive.

Future studies would benefit from a balanced experimental design in which all biological conditions are represented across multiple batches. Such an approach would greatly improve statistical robustness and enable more confident separation of technical and biological sources of variation.

4.4 Incomplete c-Jun Knockout

Validation of the c-Jun knockout revealed substantial residual JUN protein abundance, ranging from approximately 30% to 64% of wild-type levels depending on the condition. This finding suggests that the knockout may not represent complete elimination of c-Jun protein and introduces an important caveat when interpreting genotype-dependent effects.

Several explanations are possible. First, the residual signal may arise from the limited peptide evidence available for JUN quantification in the DIA dataset. Second, the CRISPR-edited cell population may contain incompletely edited cells, resulting in residual protein expression. Third, c-Jun protein may exhibit sufficient stability to persist even after successful disruption of the underlying gene. The elevated residual JUN abundance observed following cisplatin treatment is particularly interesting and may reflect stress-induced stabilization of existing protein or compensatory transcriptional responses.

Regardless of the underlying mechanism, incomplete suppression of JUN could reduce the magnitude of observed genotype effects and contribute to the differences between genetic and pharmacological perturbations. Independent validation using orthogonal approaches such as western blotting, immunofluorescence, or targeted proteomics would therefore strengthen confidence in the knockout model.

4.5 Future Directions

Several avenues could extend and strengthen the findings presented here. First, future experiments should employ balanced batch allocation across all treatment groups and time points to minimize confounding effects. Second, independent validation of c-Jun knockout efficiency would help clarify the extent to which residual JUN expression influences the observed proteomic phenotypes.

Third, inclusion of additional intermediate time points would enable more detailed characterization of the temporal dynamics of the cisplatin response and help identify early versus late stress-response programs. Finally, phosphoproteomic analysis represents a particularly promising direction because JNK signaling is mediated primarily through phosphorylation events rather than changes in protein abundance. Direct measurement of phosphosite dynamics could reveal pathway activity that is not captured by global proteome profiling and provide a more mechanistic understanding of how JNK inhibition and c-Jun loss influence the cellular response to cisplatin.

5. Data Availability

- SDRF file: Team3_Spectronaut_SDRF.tsv -- validated against ms-proteomics, human, cell-lines, and dia-acquisition templates (4/4 pass)
- Analysis code: SpectroNaut_analysis.Rmd -- fully reproducible R Markdown pipeline available at https://github.com/Peyman-13/MSProt2026_Group3/tree/main/team3
- Combined FASTA: Comb_uniprotkb_2026_05_18_Contaminants.fasta -- available on GitHub
- Spectronaut report: Team3_Spectronaut_Report.tsv -- precursor-level output submitted via FileSender - Course repository: https://github.com/jafarilab/MSProt2026

6. References

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